

Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

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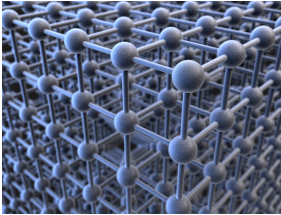
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Non-linear phenomena

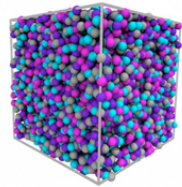
Crystal Transitions and Disorder

Crystalline



Stronger disorder
→
←
Weak disorder

Amorphous

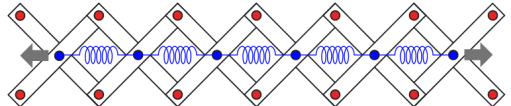


Meta-materials

spring-chain



spring-chain system embedded into a pantograph



Scientific Context: Nonlinearity & Instability

The Challenge: Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

- ▶ **Phase Transitions:** Martensitic transformations.
- ▶ **Plasticity:** Dislocation-mediated flow and avalanches.
- ▶ **Fracture:** Brittle-to-ductile transitions and damage.

Unifying Perspective

Treating these diverse phenomena as *displacive instabilities* driven by the competition between elastic energy and lattice symmetry.